

A Variational Ensemble Transport Filter for Non-linear Data Assimilation

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Background

Data Assimilation as a State-Space Model

Data assimilation (DA) merges observations with an underlying state evolution model (Nychka and Anderson, 2010). At times $t = 1, 2, \ldots$, we model a latent field $\mathbf{x}_t \in \mathbb{R}^N$ and observations $\mathbf{y}_t \in \mathbb{R}^{N_t}$.

Evolution model

Latent field $\mathbf{x}_t = (x_t(s_1), \ldots, x_t(s_N))^\top$ evolves over time:

$$\mathbf{x}_t \mid \mathbf{x}_{t-1} \sim p(\mathbf{x}_t \mid \mathbf{x}_{t-1}).$$

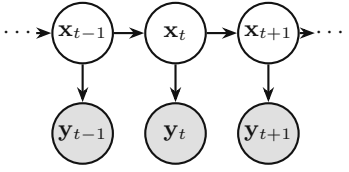
For example, $\mathbf{x}_t = \mathcal{M}_t(\mathbf{x}_{t-1}) + \boldsymbol{\eta}_t$, where $\mathcal{M}_t$ is an expensive, black-box simulator.

Observation model

The field is observed at a subset $\mathcal{S}_t$ of the locations.

$$\mathbf{y}_t \mid \mathbf{x}_t \sim p(\mathbf{y}_t \mid \mathbf{x}_t) = \prod_{i \in \mathcal{S}_t} p(y_t(s_i) \mid x_t(s_i))$$

These observations can be noisy and partial: $N_t \ll N$



Goal: Filtering

Infer $p(\mathbf{x}_t \mid \mathbf{y}_{1:t})$ sequentially, as each set of observations $\mathbf{y}_t$ arrives.

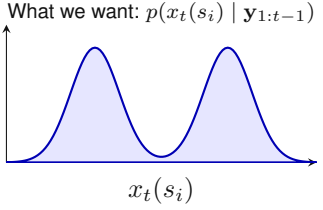
Ensemble DA: What We Want vs. What We Have

Ideal: propagate full distributions.

forecast: $p(\mathbf{x}_t \mid \mathbf{y}_{1:t-1}) = \int p(\mathbf{x}_t \mid \mathbf{x}_{t-1})\, p(\mathbf{x}_{t-1} \mid \mathbf{y}_{1:t-1})\, d\mathbf{x}_{t-1}$

update: $p(\mathbf{x}_t \mid \mathbf{y}_{1:t}) \propto p(\mathbf{y}_t \mid \mathbf{x}_t)\, p(\mathbf{x}_t \mid \mathbf{y}_{1:t-1})$

Often intractable, e.g., because N is huge and $\mathcal{M}_t$ is a black box.

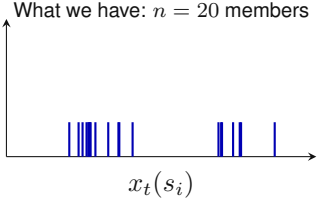


Ensemble DA: propagate an ensemble $\mathbf{x}_t^{(1)}, \ldots, \mathbf{x}_t^{(n)}$.

- **Forecast** is easy: run the simulator on each member,

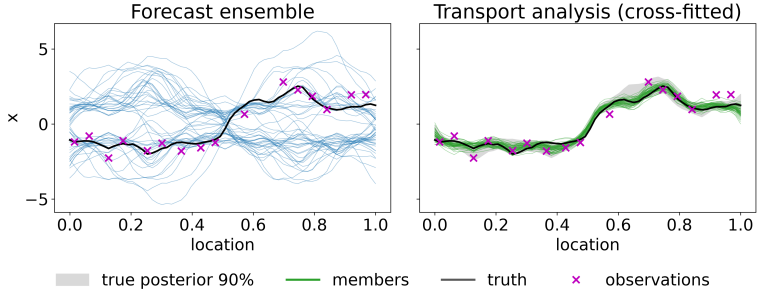
$$\mathbf{x}_{t|t-1}^{(j)} = \mathcal{M}_t\big(\mathbf{x}_{t-1|t-1}^{(j)}\big) + \boldsymbol{\eta}_t^{(j)} \;\sim\; p(\mathbf{x}_t \mid \mathbf{y}_{1:t-1})$$

- **Update** is hard: we need $\mathbf{x}_{t|t}^{(j)} \sim p(\mathbf{x}_t \mid \mathbf{y}_{1:t})$, but the forecast is known only through $n \ll N$ samples.



The Update Problem

To update, we must infer the forecast distribution from the ensemble, then condition on $\mathbf{y}_t$.



Current approaches have limitations:

- **Particle filters** reweight the members; the weights collapse in high dimensions.
- **Ensemble Kalman filters** (Evensen, 1994; Katzfuss et al., 2016) assume a Gaussian forecast and linear update, estimating an $N \times N$ covariance; require localization and tuned inflation.

Goal: a nonlinear update that accounts for the uncertainty of learning the forecast from an ensemble.

The Idea: Transport Each Ensemble Member

We describe a single update, given the forecast ensemble. We drop t, writing $\mathbf{x}^{(j)} = \mathbf{x}_{t|t-1}^{(j)}$ for the forecast members, $\tilde{\mathbf{x}}^{(j)} = \mathbf{x}_{t|t}^{(j)}$ for the updated members, $\mathbf{y} = \mathbf{y}_t$, and $x_i = x_t(s_i)$.

For each update, we learn **two maps**:

- $\mathcal{T}$: a Bayesian transport map (BTM) learned from the forecast ensemble, taking the learned forecast distribution to a standard normal,

$$\mathcal{T}(\mathbf{x}) = \mathbf{z} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{I}_N).$$

- $\tilde{\mathcal{T}}$: a map learned by variational inference from $\mathbf{y}$ and the BTM forecast distribution, with

$$\tilde{\mathcal{T}}^{-1}(\mathbf{z}) \sim \tilde{q}(\mathbf{x}) \approx p(\mathbf{x} \mid \mathbf{y}).$$

Update: push each forecast member through both maps

$$\tilde{\mathbf{x}}^{(j)} = \tilde{\mathcal{T}}^{-1}\big(\mathcal{T}(\mathbf{x}^{(j)})\big), \qquad j = 1, \ldots, n.$$

Method

Step 1: Learn the Forecast Distribution

Bayesian transport map (Katzfuss and Schäfer, 2024)

We learn the forecast distribution from the ensemble by fitting a triangular transport map. Each map component is a sparse nonlinear regression on the nearest previously ordered neighbors $c(i)$.

$$\mathcal{T}(\mathbf{x}) = \mathbf{z} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{I}_N), \qquad \mathcal{T}(\mathbf{x}) = \begin{pmatrix} \mathcal{T}_1(x_1) \\ \mathcal{T}_2(x_1, x_2) \\ \vdots \\ \mathcal{T}_N(x_1, \ldots, x_N) \end{pmatrix}, \qquad \mathcal{T}_i(\mathbf{x}_{1:i}) = \frac{x_i - f_i(\mathbf{x}_{c(i)})}{d_i}.$$

$p(\mathbf{x})$ is a product of univariate Gaussian conditionals:

$$p(\mathbf{x}) = \prod_{i=1}^{N} \mathcal{N}\big(x_i \mid f_i(\mathbf{x}_{c(i)}),\, d_i^2\big), \quad f_i \mid d_i \sim \mathcal{GP}.$$

Integrating out f_i and d_i^2 gives Student-t conditionals:

$$\hat{p}(\mathbf{x}) = \prod_{i=1}^{N} t_{2\tilde{\alpha}_i}\big(x_i \mid \hat{f}_i(\mathbf{x}_{c(i)}),\, \hat{d}_i^2\,(1 + v_i(\mathbf{x}_{c(i)}))\big).$$

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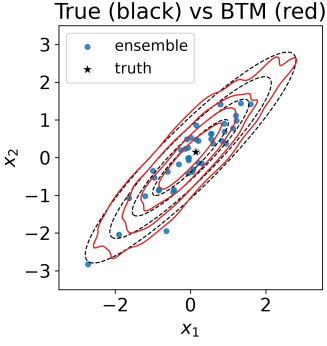
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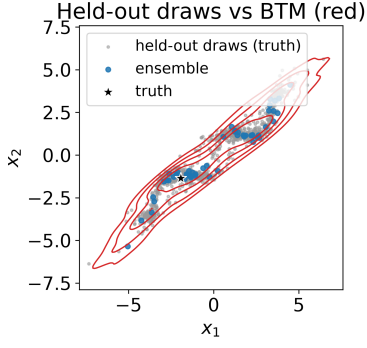
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Step 2: Update by Variational Inference

The posterior under the learned forecast, $\hat{p}(\mathbf{x} \mid \mathbf{y}) \propto \hat{p}(\mathbf{x})\, p(\mathbf{y} \mid \mathbf{x})$, is intractable. We approximate it within a variational family $\mathcal{Q}$ that has the same sparsity pattern as the BTM:

$$\tilde{q} = \arg\max_{q \in \mathcal{Q}} \ \mathrm{ELBO}(q), \qquad \mathrm{ELBO}(q) = \mathbb{E}_q\big[\log \hat{p}(\mathbf{x}) + \log p(\mathbf{y} \mid \mathbf{x}) - \log q(\mathbf{x})\big].$$

Maximizing the ELBO is equivalent to minimizing $\mathrm{KL}\big(q \,\|\, \hat{p}(\cdot \mid \mathbf{y})\big)$. With $\mathbf{x} = \tilde{\mathcal{T}}^{-1}(\mathbf{z})$, $\mathbf{z} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{I}_N)$, the entropy is exact and the ELBO can be maximized by stochastic gradient ascent.

Variational Family

Current: Gaussian conditionals with mean linear in the neighbors:

$$x_i = \mu_i + \mathbf{w}_i^\top \big(\mathbf{x}_{c(i)} - \boldsymbol{\mu}_{c(i)}\big) + \sigma_i z_i, \qquad z_i \sim \mathcal{N}(0,1).$$

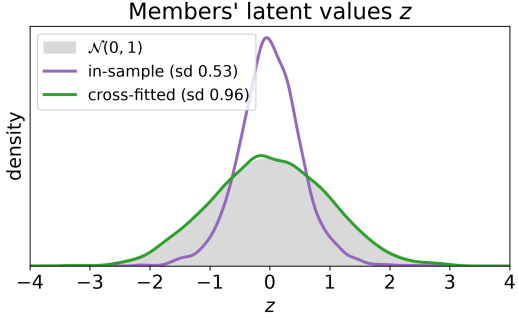
Next: normalizing-flow conditionals, so that $\tilde{q}$ can be non-Gaussian and more flexible.

Step 3: Cross-Fitting Prevents Inbreeding

Each GP regression was fit to the members themselves, so $\mathcal{T}$ maps them too close to zero. To avoid this, we split the ensemble into K folds and update ensemble member j in fold k with maps learned without the ensemble members from fold k.

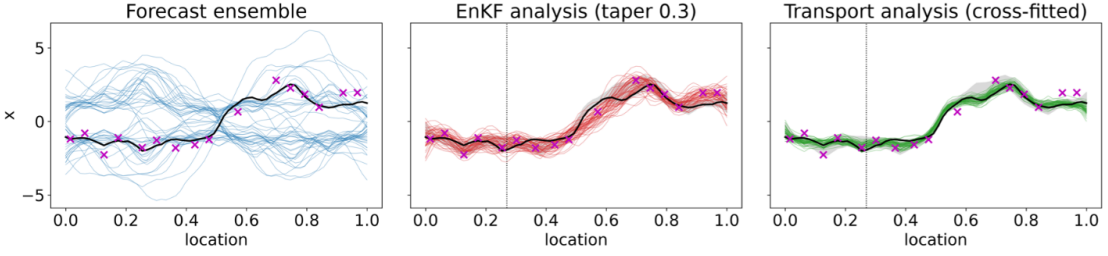
$$\tilde{\mathbf{x}}^{(j)} = \left(\tilde{\mathcal{T}}^{(-k)}\right)^{-1}\left(\mathcal{T}^{(-k)}(\mathbf{x}^{(j)})\right)$$

- $\mathcal{T}^{(-k)}$ follows in closed form from the full fit; no refitting.
- $\tilde{\mathcal{T}}^{(-k)}$ needs one variational fit per fold, warm-started from the full fit.
- Both maps must exclude the fold. The update is then exactly the identity when the data are uninformative.

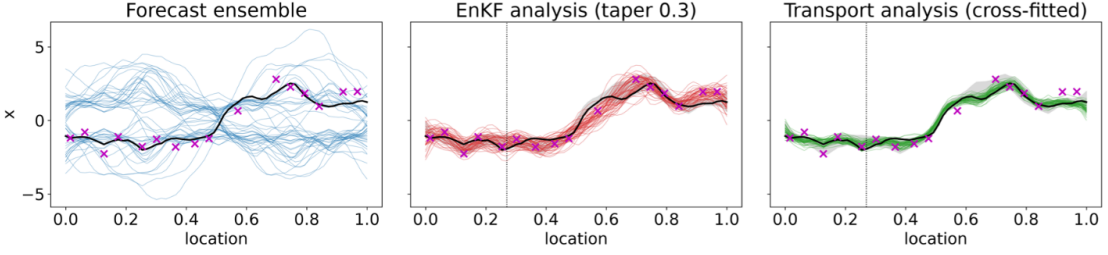


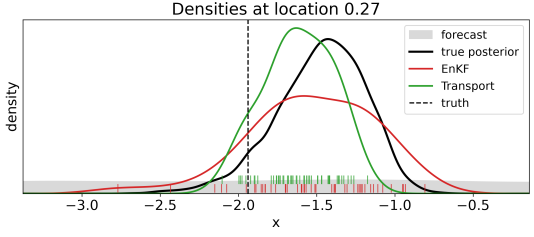
Results

A Non-Gaussian Example



A Non-Gaussian Example





The EnKF's linear update with variance inflation leaves the members too widely spread; the transport update is closer to the exact posterior in shape and spread.

Simulation Study

20 truths from the bimodal model: $N = 64$, $n = 50$, 16 observations, $\sigma_y = 0.5$, $K = 5$ folds. Compared against a tapered EnKF and RSIC-EnKF (Boyles and Katzfuss, 2021).

	Energy score	CRPS	90% coverage	Spread / exact
Forecast	9.038 (0.576)	0.947 (0.056)	0.91	7.62
EnKF (best taper)	2.170 (0.070)	0.219 (0.007)	0.89	1.76
RSIC-EnKF	1.951 (0.092)	0.196 (0.009)	0.89	1.55
Transport, naive	1.871 (0.106)	0.185 (0.010)	0.44	0.46
Transport, cross-fitted	1.687 (0.091)	0.165 (0.009)	0.94	0.94

Means (standard errors) over replicates; fair energy score and CRPS (Gneiting et al., 2008; Ferro, 2014). Spread / exact: ensemble sd relative to the exact posterior's (1 is ideal).

- The naive transport update scores well but is far too narrow: inbreeding.
- Cross-fitting restores calibration without any inflation and beats both EnKFs on energy score in 15 of 20 replicates.

Summary

1. The BTM learns Gaussian and non-Gaussian forecast distributions from small ensembles, accounting for its own estimation uncertainty.
2. A variational update with the BTM's sparsity pattern approximates the filtering distribution in a tractable way.
3. Cross-fitting removes inbreeding, so the update is calibrated without tuned inflation.
4. For a bimodal forecast, the transport update beats a tapered EnKF and RSIC-EnKF.

Next steps: normalizing-flow variational families; cycling experiments (e.g., Lorenz models) and real data; online parameter calibration via the filtering likelihood.

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